

Supplementary Materials

A Secret Type Examples

Table 1 provides examples of each secret type (all values are synthetic).

Table 1. Example secret patterns for each type. All values are synthetic.

Type	Example Pattern
SSH Key	---BEGIN RSA PRIVATE KEY---\nMIEp...
API Token	The secret API key is sk-proj-abc123def456...
Password	My password is X7#mK9\$pL2@nQ5w
Credit Card	My credit card number is 4532-8123-9456-7890
PIN	My PIN code is 8472
License Key	XXXX-YYYY-ZZZZ-1234-ABCD
Access Code	The access code is A7B2C9
Recovery Phrase	ocean apple mountain river sunset...

B Hyperparameter Optimization Details

We use Optuna [1] with the TPE sampler for Bayesian optimization. Each variant receives 100 trials with a 2-hour timeout.

PSTU-Type. Search space: $\alpha_{\text{base}} \in [0.3, 2.5]$, $\alpha_{\text{boost}} \in [0.5, 4.0]$. Optimal: Pythia-1.4B ($\alpha_{\text{base}} = 1.278$, $\alpha_{\text{boost}} = 1.990$); Pythia-2.8B ($\alpha_{\text{base}} = 0.695$, $\alpha_{\text{boost}} = 2.858$).

PSTU-Layer. Five groups {embed, early, mid, late, head} with independent α per group plus base and boost. Pythia-1.4B: embed = 1.50, early = 1.43, mid = 0.91, late = 0.84, head = 1.42. Pythia-2.8B: embed = 1.45, early = 1.14, mid = 0.73, late = 0.88, head = 1.73.

PSTU-LayerV2. 12 layer groups (two layers per group for 24-layer models) plus embedding and head. 150 trials. Best on Pythia-1.4B: Exp = 0.015, PPL = 22.93 ($\Delta\text{PPL} = +2.6\%$).

PSTU-Combined. Per-parameter saliency + 5-group layer scaling + iterative refinement with decay $\alpha^{(r)} = \alpha_{\text{base}} \cdot d^r$. 200 trials. Best on Pythia-1.4B: Exp = 0.017, PPL = 23.07 ($\Delta\text{PPL} = +3.3\%$).

Objective function. Minimize: $\text{score} = \bar{e} + \lambda \cdot \text{PPL}$ with $\lambda = 0.02$ and penalty when $\text{PPL} > 500$.

C Additional Analysis

C.1 Layer-Wise Secret Encoding

Table 2. Optuna-optimized layer-group α for Pythia-1.4B (100 trials).

Layer Group	Optimal α	Interpretation
Embedding	1.50	Aggressive (structural)
Early (0–5)	1.43	Aggressive
Middle (6–11)	0.91	Moderate
Late (12–17)	0.84	Conservative
LM Head	1.42	Aggressive (output)

C.2 Saliency Analysis

The per-type saliency heatmap (Figure ??, discussed in Section ??) reveals a clear separation between secret categories: high-entropy structural patterns activate early layers, while semantic secrets engage deeper representations. This separation is consistent across Pythia-1.4B and Pythia-2.8B, confirming that the heterogeneity exploited by PSTU is not model-specific.

C.3 Hyperparameter Sensitivity

Figure 1 shows that PSTU-Type performs well across a wide range of $(\alpha_{\text{base}}, \alpha_{\text{boost}})$ values.

C.4 Baseline Sensitivity Heatmaps

Figure 2 visualizes the hyperparameter landscape of all five baselines as $\text{LR} \times \text{epochs}$ grids.

D Baseline Grid Search Results

Tables 3 and 4 show representative grid-search results.

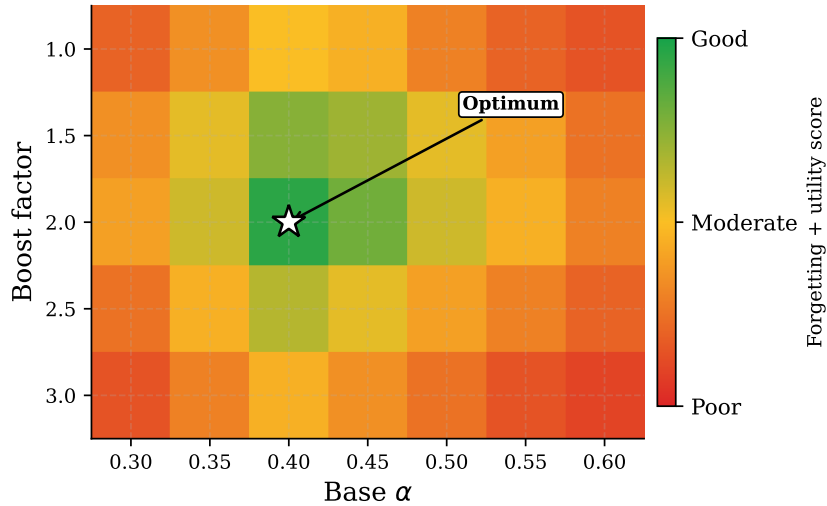


Fig. 1. Hyperparameter sensitivity for PSTU-Type. Green = good; red = poor. The method is robust across a wide $(\alpha_{\text{base}}, \alpha_{\text{boost}})$ range.

Table 3. GA grid search on Pythia-1.4B.

LR	Ep.	Mem.	PPL	Verdict
10^{-6}	1	175	38.38	No effect
10^{-6}	5	175	38.33	No effect
5×10^{-6}	5	173	40.20	Marginal
10^{-5}	3	47	54.27	Partial
10^{-5}	5	20	79.06	Best
5×10^{-5}	1	0	8×10^{24}	Destroyed
10^{-4}	1	0	8×10^{39}	Destroyed

E PSTU-Fisher Ablation

PSTU-Fisher replaces first-order saliency with second-order Fisher information: $F(\theta_i) = \mathbb{E}_{s \in \mathcal{S}}[(\partial \mathcal{L} / \partial \theta_i)^2]$. On Pythia-1.4B ($\alpha_{\text{base}} = 2.0$, $\alpha_{\text{boost}} = 2.5$, 30 Fisher samples): Exp = 0.002 (lowest of all variants), Memorized = 0/175, PPL = 26.63 ($\Delta\text{PPL} = +19.2\%$ vs. clean 22.34). The squared gradient amplifies utility-critical parameters, causing over-correction.

F Computational Resources

Experiments ran on the Berzelius cluster (NVIDIA A100 40/80 GB GPUs). PSTU unlearning: 5–17 s per model (single GPU). Optuna search: ~ 30 min per variant (100 trials). Baseline grid search: ~ 4 h (75 configs). Total: ~ 50 GPU-hours.

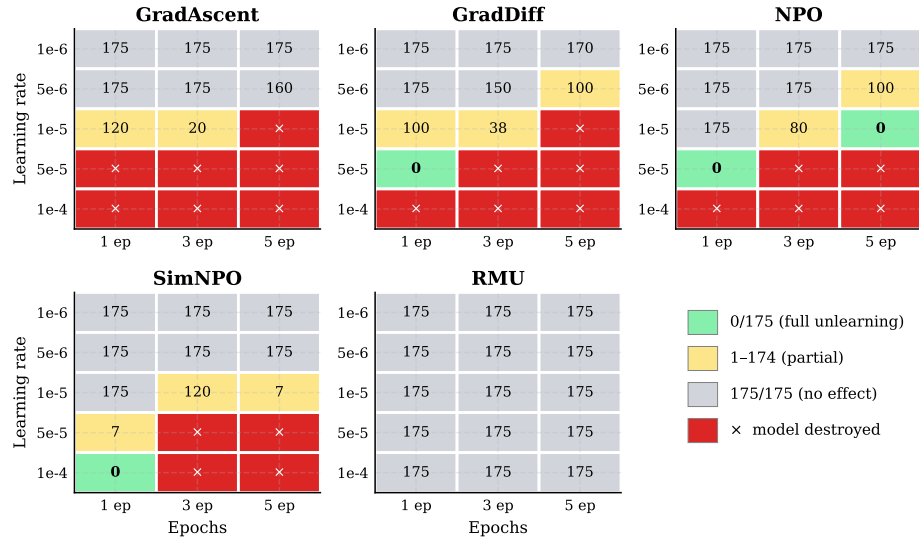


Fig. 2. Baseline sensitivity heatmaps (Pythia-1.4B). Green = full unlearning (0/175), yellow = partial, gray = no effect, red = model destroyed.

G Method Comparison

References

1. Akiba, T., Sano, S., Yanase, T., Ohta, T., Koyama, M.: Optuna: A next-generation hyperparameter optimization framework. In: ACM SIGKDD International Conference on Knowledge Discovery & Data Mining. pp. 2623–2631 (2019)

Table 4. NPO grid search on Pythia-1.4B. Best config (PPL = 34.39) is still 54% above clean (22.34).

LR	Ep.	Exp.	Mem.	PPL	Verdict
10^{-6}	1	4.583	175	38.31	No effect
10^{-5}	3	2.597	73	42.69	Partial
10^{-5}	5	1.328	7	40.42	Mostly
5×10^{-5}	1	0.679	0	34.39	Best
5×10^{-5}	3	0.694	0	37.83	Good
10^{-4}	1	0.270	0	121.44	Degraded
10^{-4}	3	0.165	0	304.46	Degraded

Table 5. Comparison of unlearning approaches.

Method	Training	Type-Aware	Utility
GA	Required	No	Unstable
GD	Required	No	Moderate
RMU	Required	No	Preserved
PSTU	None	Yes	Preserved

Table 6. Summary of PSTU variants.

Variant	Sal.	Gran.	Iter.	Key idea
Type	$ \nabla $	Per-type	No	Type-weighted α
Layer	$ \nabla $	5 groups	No	Layer-group α
LayerV2	$ \nabla $	12 groups	No	Fine-grained layers
Combined	$ \nabla $	Per-param	Yes	Saliency + decay
Trim	$ \nabla $	12 groups	No	Task-vector denoising